

Technology Stack

Date	1 October 2026
Team ID	6ab296c9dbd693aad5a6021
Project Name	FARM MANAGEMENT SYSTEM
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., HTML, CSS, JavaScript)	HTML, CSS, JavaScript	Provides a responsive and user-friendly interface for managing crops, farmers, livestock and farm activities.

2	Backend / Server-Side (e.g., Python/Django, Java)	Python / Django	Handles secure business logic, user management and farm-data processing.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	MySQL	Provides structured storage for farmer, crop, livestock, inventory and farm records.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Render, Docker)	Render	Offers simple and cost-effective deployment with scalability.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub	Enables team collaboration, version control and reliable code management.
6	Third-Party APIs / Other Tools (e.g., Weather API, Stripe, Twilio, Google Maps)	Weather API	Provides weather forecasts useful for crop planning and farm decisions.